



OPEN Rubber-tracked crawler undercarriage multibody dynamic simulation for agriculture operation

Ramiro Jose Chamorro Coneo^{1✉}, Wesllen Lins de Araujo¹, Angel Pontin Garcia², Marcelo Vinícius de Paula¹ & Auteliano Antunes dos Santos¹

This paper presents a novel multibody dynamic model for agricultural vehicles and applies it to study the effects of dynamic loading on a tracked wheel crawler vehicle used for agricultural purposes. Multibody dynamic simulation (MBS) is used to analyze the vehicle's behavior under various operating conditions, considering the stiffness between tracks and the arrangement of a rubber track system. The model accounts for soil interaction with the crawler and the load effects on different soil types, resulting in slip ratios up to 9% for dry sand, 8% sandy loam, and 6% clay soil. The rubber crawler vehicle is simulated in different scenarios, including a 20° slope plane climb and a draft load of 13 kN representing a planter and with seeding equipment. Under these conditions, the model reveals that the slip ratio increases to 20% in certain conditions. The results also show that the tension in the crawler belt element increases until it reaches 7.5 kN due to the type of soil on the bottom rollers when the vehicle has no draft load operation. The approach adopted here is a promising way to design better agricultural machinery optimized for efficiency and sustainability.

Keywords MBS, Tracked wheel vehicle, Flexible tracks, Agricultural vehicle modeling

The crawler for agricultural applications exhibits a complex interaction between mechanical components (rollers/tracks), and the terrain plays a fundamental role in determining the machine's performance¹. There are two main types of crawlers: rigid and flexible tracks. These types are related to their structure, material, and behavior. Rigid tracks consist of steel plates connected by pins or hinges, forming a solid chain that wraps around the drive and road wheels. While rigid tracks offer high strength and durability, they also come with significant weight and stiffness, which can lead to elevated impact forces and stresses on both the vehicle components and the ground. A flexible tracked wheel vehicle is an off-road vehicle that utilizes continuous tracks instead of tires to navigate uneven terrain. The track belt is made of rubber reinforced with metal cords and or metal inserts connected by hinges or joints, allowing it to flex and adapt to the soil. Rubber crawlers offer several advantages over metal tracks, including quieter operation, longer service life, lower operating costs, and improved traction. However, the internal motion resistance (associated with the internal components of the crawler) can be high, resulting in increased energy consumption. One of the factors that affect internal resistance to movement and can reduce vehicle efficiency is the proper control of the track tension. In principle, increasing track tension helps keep them in place on difficult or rocky terrain and reduces friction. However, keeping them too tight increases track wear, both for rubber and steel tracks. This can also be affected by the vehicle's operating load, which impacts the vehicle's slippage and reduces the energy efficiency of the planting operation. The magnitude of the forces and their relation cannot be determined during the design process and is very difficult to measure in focused and comprehensive prototypes. So, analytical or numerical analysis is vital. Because of that, it is essential to examine the stiffness and damping properties of rubber tracks².

One way to estimate the forces and vehicle behavior is through multibody simulation models or MBS models. MBS is a numerical method that allows the analysis of motion and interactions in complex systems composed of multiple rigid or flexible bodies. In recent years, due to the development of computational resources, numerical simulation has become an essential tool to predict the behavior of complex mechanical systems³. These simulations enable the analysis of different operating conditions of a mechanical system during the early design stage. Multibody models are often used to describe the kinematic and dynamic behavior of vehicles. MBS has been widely used for crawler vehicles, especially in the case of rigid tracks, especially in mining^{4,5}, military^{6,7}, and agricultural applications^{8–10}.

¹Faculdade de Engenharia Mecânica (FEM), Universidade Estadual de Campinas, Campinas, SP 13083-860, Brazil.

²Faculdade de Engenharia Agrícola (FEAGRI), Universidade Estadual de Campinas, Campinas, SP 13083-875, Brazil.

✉email: rccone@unicamp.br

For agricultural purposes, the MBS should account for sinkage pressure, shear stress, vehicle slip ratio and uneven terrain, in order to model the vehicle dynamic behavior. Researchers such as Grazioso et al.¹¹ developed an MBS model in MATLAB, focusing on a half-agricultural vehicle. The study aims to optimize the distribution of ground pressure and minimize soil damage for a small agricultural vehicle equipped with wheels and tracks. A detailed multibody model was created to evaluate the pressure distribution of the flexible track of the vehicle in soft terrain. The track module includes a roller chain track, sprockets, and a reconfiguration mechanism designed to adapt to different soil conditions. Parameters such as chain tension, shock absorber stiffness, and idle preload are considered in the MBS design. Although MBS was initially developed for rigid tracks, rubber shoes are considered.

Despite that work, MBS models for rubber tracks are relatively uncommon, with a few exceptions. Authors like Mocera and Nicolini¹² developed an MBS software (ADAMS®) model for a small agricultural track vehicle. The detailed model of each track segment was combined with a rigid body model of the vehicle. The authors analyzed the relationship between the track and the underlying surface on deformable terrain, specific forces were meticulously assigned to each component of the track model. These force definitions were derived from deformable soil mechanics equations, and implemented using specially designed routines within the multibody code. According to the proposed model, the primary terrain deformation is assumed to be concentrated around the vehicle's track elements. A shear ground stress model was considered to describe the behavior of the terrain under the influence of the crawler. The study focused on the results related to traction, slippage, and subsidence on different types of terrain. Tests conducted in a multibody environment demonstrate how the proposed model can achieve tensile performance similar to the corresponding analytical solution. Additionally, the web plate enhances the availability of tensile forces for specific soil properties.

Later, Nicolini et al.¹³ and Mocera et al.¹⁴ improved their model by considering grousers on the tracks and developing multibody simulation (MBS) models that incorporate contact equations accounting for soil deformation due to tracked wheel vehicle interaction. While the MBS model initially assumes that the action of rigid sprocket teeth is on the track belt, this may not hold true for rubber crawlers. Additionally, although they compared traction pull bars with and without grousers during slope climbing, the model was not tested under draft load.

Other authors like T. Fukushima et al.¹⁵, developed an MBS model to analyze rubber track tension oscillation and evaluated the relationship between crawler tension and driving performance. This rubber track crawler model can simulate fluctuations in crawler tension, considering parameters like stiffness and damping in agricultural vehicles. The model analysis concluded that the elastic parameters of the crawler track band and idle support spring significantly affect driving performance, including crawler tension, driving torque, and traveling speed. The MBS was developed in RecurDyn for a half vehicle with three rollers, two idler wheels, and a sprocket. It simulates the tracks connecting with stiffness and damping elements while considering the flexibility between each track. However, the model was limited to ground conditions, considering the same pressure along the tracks and using a simple Coulomb friction model. Soft soil, such as that found in agricultural applications, can generate nonhomogeneous vertical load distribution depending on the soil condition, as described by other authors^{16,17}. Another part that was left out of this analysis was the vehicle's draft force and its potential consequences due to drive torque on track belt tension.

On the other hand, some authors have investigated how vehicles perform under various scenarios, aimed at improving their designs. Mou et al.¹⁸ evaluated the possibility of a tracked universal chassis in hilly and mountainous orchards. The study focused on three key working conditions: climbing, crossing obstacles, and traversing trenches. Exploring the theoretical maximum climbing angle, maximum obstacle height, and maximum trench width through kinematic calculations. The chassis model's passability was then simulated using RecurDyn dynamics software. Field tests validated the results, demonstrating that the crawler could smoothly traverse slopes with a 35° angle, climb vertical obstacles up to 200 mm in height, and pass through trenches with a width of 430 mm. While the model establishes parameters for soft soil, it does not specifically address vehicle stability or provide details about loads and tensions in the track belt.

This article aims to model a rubber-tracked vehicle under standard loads for seeding and planting operations across different soil types. This MBS model of the crawler uses internal contact based on the specific geometry of the components in contact and their materials, employing the polygonal contact model (PCM). Similar models use line contacts based on constrained planar movements, which in turn restrict certain types of movements between the elements of the vehicle. On the external part, which is in contact with the ground, a nonlinear viscoelastic model is used, incorporating damping as an important detail in the dynamic analysis of the soil. The contact system is divided into track segments. The model also allows for the analysis of shear forces with sliding and the resistance to movement generated by the soil, which enables an assessment of the behavior of the tension in the rubber belt. The model takes into account the effects of three soil types: dry sand, clay loam, and clay soil. Specifically, the model accounts for sinkage forces at the track/soil contact, motion resistance, and shear forces generated when the vehicle interacts with the soil. These features represent an advance in the state of art on modelling this type of vehicle, when comparing with the references listed. The track system is represented to consider the tension between links, allowing the model to consider not only soil conditions but also belt tension as part of the simulation results. This study also highlights the importance of factors such as belt tension because it allows the determination of the force variations that occur in the belt, given the changes in different soils and operating conditions. In the following section, a series of simulations are used to verify the results and assess the model's efficiency.

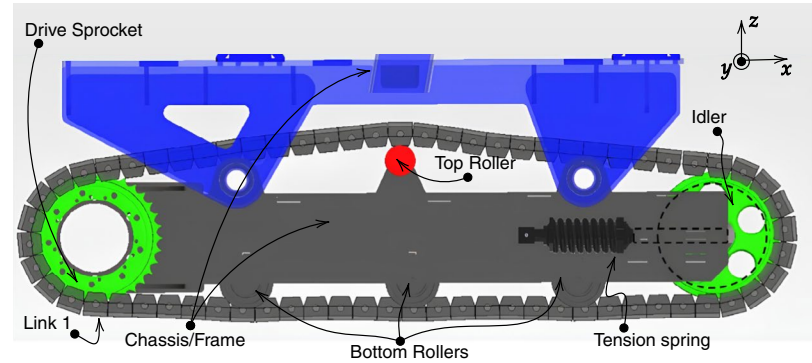


Fig. 1. The crawler system used during the simulations.

Geometric parameters	Values [m]
Sprocket diameter	0.350
Idler diameter	0.330
Total track length	2.1
Distance between sprocket and idler center	1.61
Diameter bottom roller	0.14
Diameter top roller	0.08
Track segment length	0.06
Track width	0.18

Table 1. Geometric characteristics of the crawler.

Methods
Multibody model description

The rubber crawler Yanmar VI027 is modeled in Simpack®, with geometric and inertial properties configured based on a real vehicle reference. The basic dimensions are given in the manufacturer site (<https://www.yanmar.com/br>). All simulations were developed and run on Windows 11 Pro using a 12th Generation Intel Core i7-1255U CPU. Animations of the simulations are available in supplementary sources. The multibody model was developed as a three-dimensional model. For our study, we considered that there are no lateral restrictions. However, the analysis for this study was conducted focusing on vertical and longitudinal forces. Lateral forces were only added to support the longitudinal movement of the vehicle. Figure 1 provides a description of each crawler part.

The vehicle is 2.1 m long, approximately 3.13 m wide, and has a height of 1.8 m. Considering the reference point at the midpoint of the vehicle’s length, the center of gravity is shifted 0.08 m toward the idler wheel, 1.09 m in height from the lower part of the track, and positioned at the midpoint of the width. The primary components of the crawler include the drive sprocket wheel, idler wheel, three bottom rollers, a top roller, the tension spring, the crawler frame, and the tracks. The main geometric characteristics of the track components are described in Table 1. The system is composed of 149 bodies, which together have 293 degrees of freedom.

The sprocket wheel drives movement by transmitting power to the tracks. In this case, the sprocket has 34 teeth for double-pitch engagement with the crawler, and it is configured to rotate in relation to the frame. The frame is connected to the idler wheel support via a spring that regulates the track tension and additionally restricts the idler support so that it only moves longitudinally. It is simulated as a unidirectional (longitudinal) spring-damping system with a preload between the frame and the support. The support is connected to the idler wheel via a bearing that restricts its movement, allowing rotation only along the y-axis. In addition, there are three bottom rollers and a top roller that rotates relative to the frame. Table 2 represents the inertial parameters with respect to the center of mass of the respective body considered for the model obtained from Simpack®. In this work x is horizontal, z is vertical, and y is perpendicular to the plane of the Figure 1

The crawler system model also considers the stiffness between each track element, which arises from the combination of rubber and steel cords. In the simulation, this stiffness is represented as a bush element placed between the tracks, accounting for both stiffness and damping at the joint. By configuring these elements, the model can accurately calculate the tension force within the crawler. Two degrees of freedom were considered between each element. For each one, an associated stiffness and damping were considered. k_x and c_x represent the linear stiffness and damping of the rubber belt, while k_β and c_β represent the angular stiffness and damping between each link². Figure 2 shows the force element, which represents stiffness and damping between every track in the crawler. The model also constrains the lateral and vertical displacement and the rotation in relation to the next track element.

Body	Characteristics [S.I]	Value	Body	Characteristics [S.I]	Value
Track frame and planter	Mass [kg]	4600	Top roller	Mass [kg]	3.66
	Inertia x-x [kg·m ²]	9207.4		Inertia x-x [kg·m ²]	0.0045
	Inertia y-y [kg·m ²]	2409.81		Inertia y-y [kg·m ²]	0.0028
	Inertia z-z [kg·m ²]	8867.03		Inertia z-z [kg·m ²]	0.0045
Sprocket	Mass [kg]	8.95	Bottom Roller	Mass [kg]	8.26
	Inertia x-x [kg·m ²]	0.09		Inertia x-x [kg·m ²]	0.022
	Inertia y-y [kg·m ²]	0.17		Inertia y-y [kg·m ²]	0.012
	Inertia z-z [kg·m ²]	0.09		Inertia z-z [kg·m ²]	0.022
Idler	Mass [kg]	17	Track segment	Mass [kg]	2.5
	Inertia x-x [kg·m ²]	0.13		Inertia x-x [kg·m ²]	0.007
	Inertia y-y [kg·m ²]	0.26		Inertia y-y [kg·m ²]	0.001
	Inertia z-z [kg·m ²]	0.26		Inertia z-z [kg·m ²]	0.007
Idler support and spring	Mass [kg]	6.3			
	Inertia x-x [kg·m ²]	0.028			
	Inertia y-y [kg·m ²]	0.044			
	Inertia z-z [kg·m ²]	0.066			

Table 2. Inertial parameters of crawler bodies used for MBS.

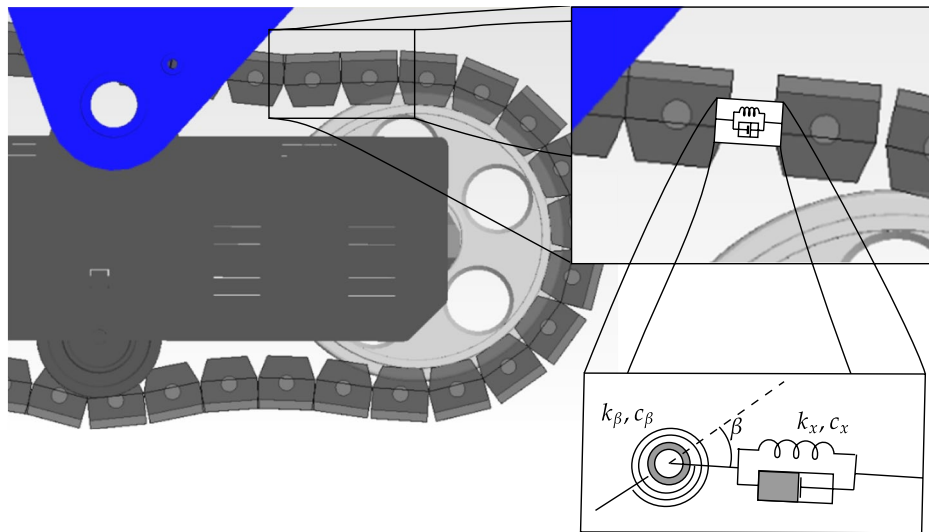


Fig. 2. Scheme of the bushing element in the tracks systems.

Contact between crawler elements

The current model used a polygonal contact model (PCM) for contact among the sprocket and the tracks, the idler and the tracks, and the rollers (bottom and top) and tracks. The PCM is a contact algorithm used by MBS to evaluate complexly shaped bodies. The contact used its polygonal representation surface of a body geometry to determine the characteristics of an elastic foundation model¹⁹. This enables the definition of contact between surfaces of nearly any shape and facilitates the calculation of multi-point contacts. At first, the PCM algorithm covers the contact bodies with a thin elastic layer, considering that the layer does not promote deformation or stress in the tangential direction. The contact path is discretized using the contact elements. Assuming that the normal pressure is constant for the combined elastic layer, then the normal direction and the area of the base polygon represent a normal pressure subdivision. The elastic force law of an element k is defined by the Equation (1)

$$F_{cl} = c_l A_k u_{nk} \quad (1)$$

where c_l is the layer stiffness, A_k is the contact element area, and u_{nk} the penetration, which is always perpendicular to the surface. For the elastic model foundation, the c_l of each body is defined by

$$c_l = \frac{K_l}{b} \quad (2)$$

where b the layer thickness and K_l is defined as

$$K_l = \frac{1 - \nu}{(1 + \nu)(1 - 2\nu)} E \quad (3)$$

In Equation (3), E is the Young's modulus and ν is the Poisson's ratio. Although the layer thickness b can be sufficient to define the elastic properties of the contact, in most applications where both bodies are similar, there is no comparatively soft layer on the body surface, so a suitable value can be hard to find. However, as b acts like a scaling factor for the layer stiffness and has a great influence on it, authors²⁰ suggest that b may be tuned by using experiments. So, given two surfaces, A and B , the total contact stiffness is calculated as series stiffness spring as

$$C_{l,total} = \frac{c_{l,A}c_{l,B}}{c_{l,A} + c_{l,B}} \quad (4)$$

where $c_{l,A}$ and $c_{l,B}$ are the stiffness layers of A and B subsequently. A similar approach can be used to calculate the damping force in a PCM contact element. The damping force for the element is calculated by

$$F_{dl} = d_l A_k \dot{u}_{nk} \quad (5)$$

where d_l is the layer damping and $\dot{u}_{nk}$ the element normal velocity. d_l cannot be specified by material properties or geometry data, but like b , it requires to be adjusted using simulations or verified values. The normal contact force is defined as:

$$F_{nl} = \begin{cases} F_{cl} + F_{dl}, & \text{if } F_{cl} + F_{dl} > 0 \\ 0, & \text{if } F_{cl} + F_{dl} \leq 0 \end{cases} \quad (6)$$

The tangential force of the element is calculated using the normal contact force F_{nl} as

$$F_{tl} = \begin{cases} \mu F_{nl}, & v_{tl} \geq v_e \\ \mu F_{nl} \frac{v_{tl}}{v_e} \left(2 - \frac{v_{tl}}{v_e}\right), & v_{tl} < v_e \end{cases} \quad (7)$$

where μ is the friction coefficient, v_{tl} is the relative tangential velocity, and v_e is the critical velocity of static friction. Due to the stiffness of the layer and the damping, several simulations have been developed to evaluate contacts in every contour figure. This procedure is used to reduce computational effort during the main simulation. Table 3 resumes the simulation conditions of the undercarriage elements parameters used in the model.

Soil track contact model

To establish the force related to the soil behavior when compressed in the normal direction, the pressure-sinkage p , also known as the Bekker equation, is used. Bekker equation²¹ is shown in Equation (8).

$$p = \left(\frac{k_c}{w} + k_\phi \right) z^n = k_{eq} z^n \quad (8)$$

In Equation (8), z is sinkage, k_c is the cohesive modulus, k_ϕ frictional modulus²², n is the sinkage exponent, and w is the reference dimension of a circular or rectangular plate, which is the link width in this case in contact with the terrain. Due to the track behavior during the contact with the soil, damping effect was suggested in Molari et al¹. They suggested the Equation (9) for normal contact:

$$F_n = A \left(\frac{k_c}{w} + k_\phi \right) z^n = k_{eq} z^n + A c_D \dot{z} \quad (9)$$

where A was the link base area, w was the link width, and c_D is a damping coefficient. c_D in this simulation was adjusted to 100,000 Ns/m³. Figure 3 shows the Bekker force acting on the track link in the MBS model. In this case, the normal force is placed on the link bottom, expressing the force in the middle of the link length.

The model used for the contact between the track and the ground is a nonlinear viscoelastic type. The contact force depends on the sinking (elastic behavior), and when using Bekker's equations, the system reacts nonlinearly due to the exponent. Additionally, the penetration velocity of each segment generates viscous damping, which allows energy dissipation and creates hysteresis in the contact load behavior with the ground.

The soil also reacts to a shear load, which is described by the Janosi-Hanamoto equation²³. The stress-shear displacement relationship in the longitudinal direction of the track is shown in Equation (10)

$$\tau = \tau_{max} \left(1 - e^{-\frac{j}{K}} \right) \quad (10)$$

where the τ is the shear stress, j is the shear displacement of the terrain, K is the shear deformations parameter and represents when the terrain reaches the maximum shear. τ_{max} is the maximum shear stress and is defined as

Parameters	Values
Track longitudinal stiffness kN/m	2500
Track longitudinal damping Ns/m	4100
Track angular stiffness Nm/rad	160
Track angular damping Nms/rad	8
Tension spring stiffness kN/m	1000
Tension spring damping kNs/m	200
Preload tension spring N	4000
Friction coefficient between the track for rollers and idler	0.3
Friction coefficient between the track insert and the sprocket	0.01
Friction coefficient between tracks and soil (lateral only)	0.5
Track insert, sprocket, idler, and rollers Young's modulus GPa	210
Rubber track young's modulus MPa	100
Sprocket, idler, rollers, track insert poisson's ratio	0.27
Track poisson's ratio	0.4
Layer thickness for the sprocket and track insert m	0.5
Layer thickness for the roller and idler m	0.01
Layer thickness for the track inner surface m	0.032
Damping for the sprocket and track insert kNs/m ³	100000
Damping for the for the roller, idler, and track kNs/m ³	10000

Table 3. Model parameters of undercarriage elements.

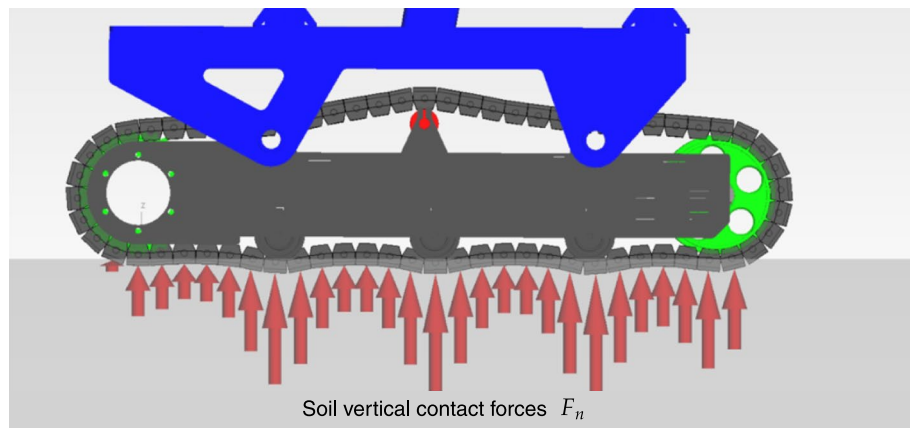


Fig. 3. The sinkage force distribution for track link.

$$\tau_{max} = c + p \cdot \tan \varphi \quad (11)$$

where p is the normal pressure (Bekker equation), c is the soil cohesion, and φ is the internal friction angle. Equation (11) is known as the Mohr-Coulomb soil failure criterion and is applied in the plastic equilibrium theory for vehicle-ground interaction. Plastic equilibrium is defined as the state that precedes plastic flow, as described by Wong²⁴. Shear displacement j can be defined as the product ix , being i expressed as:

$$i = \begin{cases} \frac{\omega r - v_x}{\omega r}, & \text{if } \omega r > v_x \text{ (wheel slip)} \\ \frac{v_x - \omega r}{v_x}, & \text{if } \omega r < v_x \text{ (wheel skids)} \end{cases} \quad (12)$$

In Equation (12), i is known as the slip coefficient, ω is the sprocket rotational speed, r is effective radio (Sprocket ratio and track height), and v_x is the vehicle velocity. This function encloses the slip coefficient between -1 and 1 interval and is only not valid for values of v_x and ω equal to 0 . From here on, x is defined as the relative longitudinal between the track segment and the center of the idler wheel, as shown in Figure 4.

Due to the shear force being the integral of Equation (10), the force for a single track can be obtained by Equation (13).

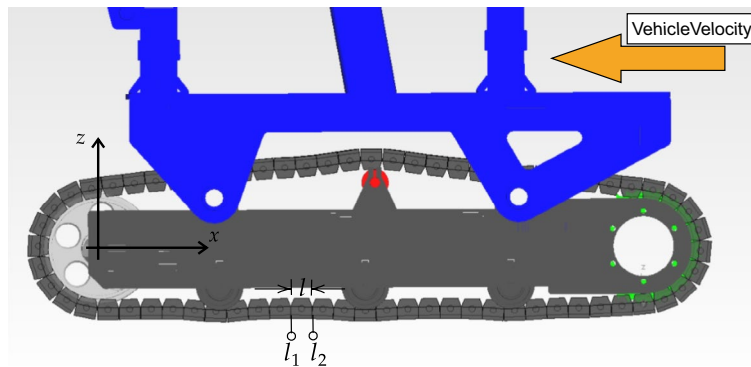


Fig. 4. Schematic of the crawler reference and limits for the integral force from Janosi-Hanamoto equation.

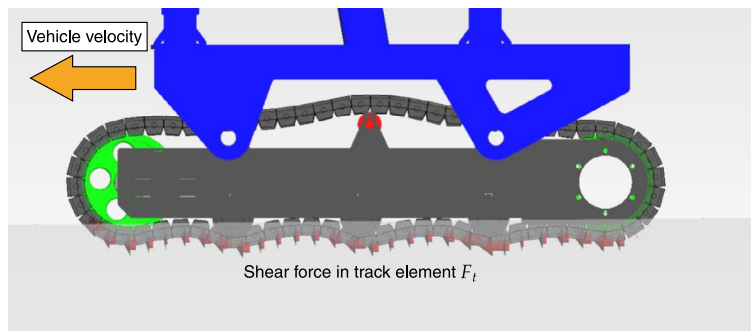


Fig. 5. Shear force distribution on track during simulation.

$$F_t = w \int_{l_1}^{l_2} (c + p \cdot \tan \varphi) \left(1 - e^{-\frac{x}{K}} \right) dx \quad (13)$$

$$= A (c + p \cdot \tan \varphi) \left[1 + \frac{K}{il} \left(e^{-\frac{il_1}{K}} - e^{-\frac{il_2}{K}} \right) \right]$$

In the previous Equation, A was the link base area, l was the track element length, and l_1 and l_2 were the limits of integration for a single track element. l_1 and l_2 are the longitudinal distance reference from the idle wheel center and calculated for every track element in contact with soil, which makes this integral moving its limit at every calculation. Figure 4 shows a scheme with the reference for the calculation of Equation (13).

The shear force is represented at the center of the track element in contact with soil in the longitudinal direction, as is shown in Figure 5.

Bekker soil parameter

For the development of the model, three different types of soil were considered: terrain with low cohesion and high friction angle, such as dry sand; terrain with higher cohesion, such as clay soil, with low friction angle; and finally, a combination of both, similar to what is found in sandy loam. The dynamics of the force produced by each of these soils will depend on both the weight of the vehicle and the contact area of the track. In the case of low cohesion and a high friction angle, the shear load will depend on the vehicle's weight. Thus, if the soil has high cohesion and a low friction angle, it will depend more on the contact area of the track. Table 4 presents the values suggested by Yong, R. N. et al.²⁵ and Wong²⁴. Additionally, the damping values of the soil were adjusted for the model. Authors such as Nicolini suggest some values that served as a starting point for adjusting this parameter. The MBS model is configured with soil types such as dry sand, sandy loam, and clay soil. Table 4 shows the values used for the soil track contacts interaction.

Motion resistance force

Motion resistance is normally calculated using the work due to deformation caused by the soil. For this, it is necessary to calculate the pressure distribution in the contact area under the rubber belt. The Equation is as follows:

$$R_c L = w \int_0^L \int_0^{z(x)} p(z(x)) dz dx \quad (14)$$

Soil parameter [S.I]	Dry sand	Sandy loam	Clay soil
n [—]	1.1	0.7	0.5
k_c [kN/m ^{$n+1$}]	0.9	5.3	13.2
k_ϕ [kN/m ^{$n+2$}]	1523.4	1515	692.2
c [kN/m ²]	1	1.7	4.14
K [cm]	2.5	2.5	1
φ [°]	30	29	13
k_{eq} [N/m ^{n}]	16506.72	16680	8267.76

Table 4. Parameter for simulation soil analyzed^{24,25}.

Where R_c , is the motion resistance force, $p(z(x))$ is the soil pressure distribution, and L is the crawler total length. It is usually assumed that the pressure on the soil is uniform along the entire length of the line, simplifying the equation²⁴. However, this is not the case. Because there are different pressure distributions, it is necessary to calculate the integral numerically. Using the vehicle's MBS, it is possible to calculate the pressure distribution of each soil using the parameters from Bekker's pressure-sinkage equation and determine the motion resistance in each case as:

$$R_c = \frac{w}{L} \int_0^L \int_0^{z(x)} \left(\frac{k_c}{w} + k_\phi \right) z^n(x) dz dx \quad (15)$$

$$= \frac{w}{L} \int_0^L \left(\frac{k_c}{w} + k_\phi \right) \frac{z^{n+1}(x)}{n+1} dx$$

In this case, the vehicle's center of gravity alters the distribution of soil pressure since it is slightly shifted toward the idler wheel. Additionally, soil pressure distributes the load across each of the track segments according to Bekker's equation, depending on its sinking. Due to this, the distribution function of $p(z(x))$ may not be a trivial integral, which makes numerical integration solutions necessary. Approximating by the trapezoidal method and using the same reference from Figure 4, the resulting equation used for the total motion resistance is as follows:

$$R_{cTotal} = \frac{wk_{eq}}{L} \left[(l_2 - l_1) \left(\frac{z^{n+1}(l_1) + z^{n+1}(l_2)}{2(n+1)} \right) + (l_3 - l_2) \left(\frac{z^{n+1}(l_2) + z^{n+1}(l_3)}{2(n+1)} \right) + \dots \right] \quad (16)$$

This approximates the value of each segment to each value as:

$$R_c = \frac{wk_{eq}}{L} (l_2 - l_1) \left(\frac{z^{n+1}(l_1) + z^{n+1}(l_2)}{2(n+1)} \right) \quad (17)$$

The equation of the segments is conditioned only to segments that are in contact with the soil and is always opposite to the movement of the vehicle as shown in Figure 6. In the case of zero velocity, the force is zero.

Work operation: vehicle at rest, vehicle with and without draft force, and vehicles on a ramp

This study aims to simulate the MBS vehicle model implemented for traction for seed operations machinery, contemplating four scenarios: Vehicle at rest, vehicle in motion without draft load, vehicle with a draft load (planting) and tilled soil without draft load.

First, the vehicle model is simulated to evaluate the behavior of the MBS of the track in contact with the three types of soil. It starts at rest and then stabilizes. This allows for the analysis of the vertical load distribution generated by soil sinkage, as well as the evaluation of the maximum motion resistance.

Subsequently, simulations are conducted with the tracks in motion. To do this, the simulation begins with the vehicle at zero speed. After that, the sprocket wheels are actuated with a controlled angular velocity profile. For this, a motor torque is required, which is adjusted to match the angular velocity. The velocity profile transitions over 1 second until reaching a constant speed of 11.11 rad/s. The angular velocity value is associated with the theoretical speed of the vehicle, based on the effective radius of the drive wheel plus the track thickness 0.2099 m, resulting in a theoretical speed of 2.33 m/s, which means the planter is running at 7 km/h. This dynamic simulation enables the calculation of target parameters such as the slip rate, shear forces, and track stresses for each soil type analyzed in this study.

Afterward, the MBS model configured a draft force in the longitudinal direction and simulated the draft produced by the seeding implement. Figure 7 shows the direction of the force draft in the MBS model scheme. Normally, the drag force is located a few centimeters below the vehicle, generated by the interaction between the planting tools and the soil. To simulate this load, it was necessary to position the drag force at the center of gravity and compensate for the torque produced in order to replicate the effect of the planter tools on the soil at a depth of 7 cm.

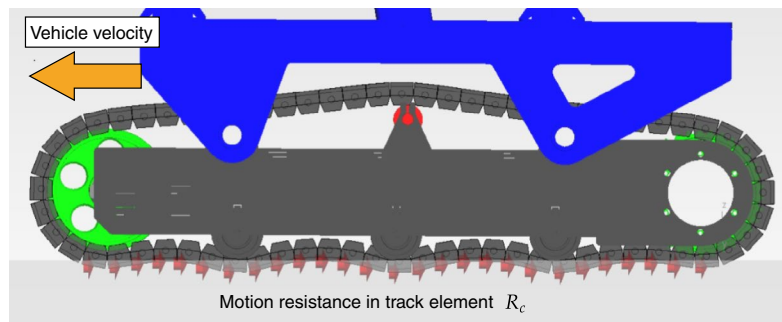


Fig. 6. Motion resistance distribution on the track during the simulation.

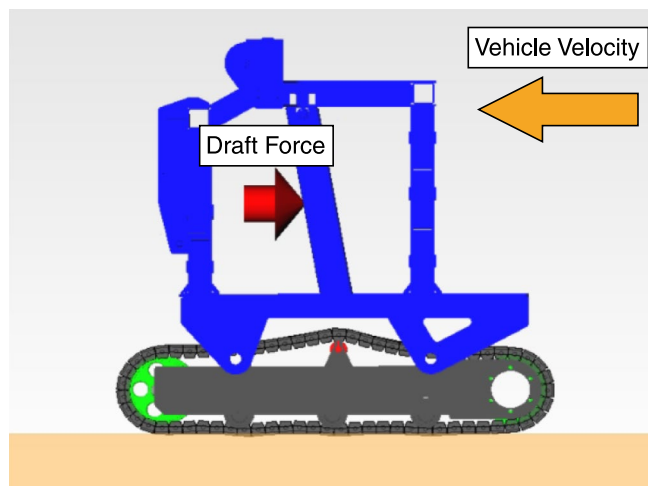


Fig. 7. MBS model indication of the draft force used in the simulation.

The draft force can be influenced by multiple factors, like the type of machinery, the number of lines of the tillage tools, the speed at which it is pulled, depths, and soil texture, among others. Considering ASAE D497.5²⁶ suggests the Equation (18) to calculate the draft force Df .

$$Df = F_i (A_{Df} + B_{Df}v_x + C_{Df}v_x^2) WT \quad (18)$$

In Equation (18), F_i is a dimensionless soil texture adjustment parameter, subindex i defines three types of textures soil (1 fine, 2 medium, and 3 coarse), A_{Df} , B_{Df} , and C_{Df} are machine specific parameters, v_x is field speed, W number of rows or tools, and T is the tillage depth. The parameters on the model are configured to 5 rows in the planter. The vehicle is adjusted for a longitudinal velocity of 7 km/h under the three different soil textures. The ASAE D497.5 for a row crop planter, zone-till, the parameters are: A_{Df} is 3400, B_{Df} and C_{Df} are 0, T is 1 since it is a planting implement, and F_1 is 1, F_2 is 0.94, and F_3 is 0.79 for soil textures and W is 5 due to there are 5 rows. This means that the draft load varies between 17,000 N and 13,430 N. For simulation, the applied drag load will gradually change with the speed of the vehicle until it reaches a constant velocity. In other words, a transition will follow speed so that when a velocity of 7 km/h is reached, the maximum load is achieved, which in this case will be 17,000 N.

Under the final working conditions, the vehicle starts in a manner similar to the scenario without a draft load. However, after reaching a speed of 7 km/h, the vehicle arrives at the beginning of the ramp with 20° inclination. During this scenario, the vehicle does not have any external force due to the planter, that is, the only force considered is due to the weight and slope as disturbances in the model. Throughout this movement, the angular velocity of the sprocket wheels is maintained. Figure 8 shows the slope in the soil condition scheme.

Results

The results for the four scenarios under evaluation are listed in the following topics.

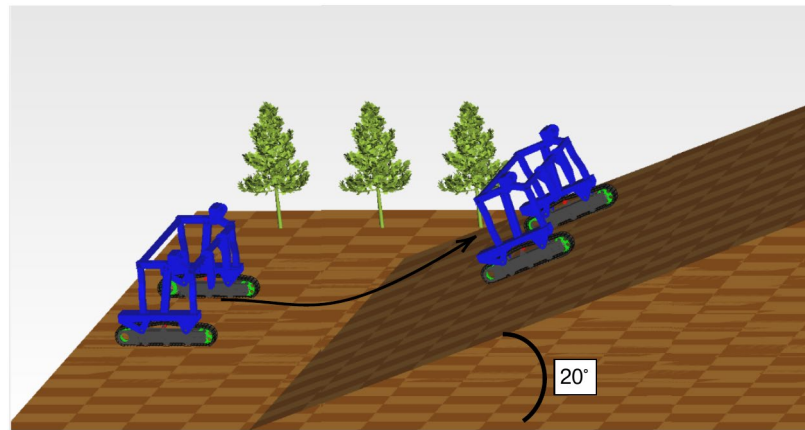


Fig. 8. Scheme of the MBS motion on the ramp.

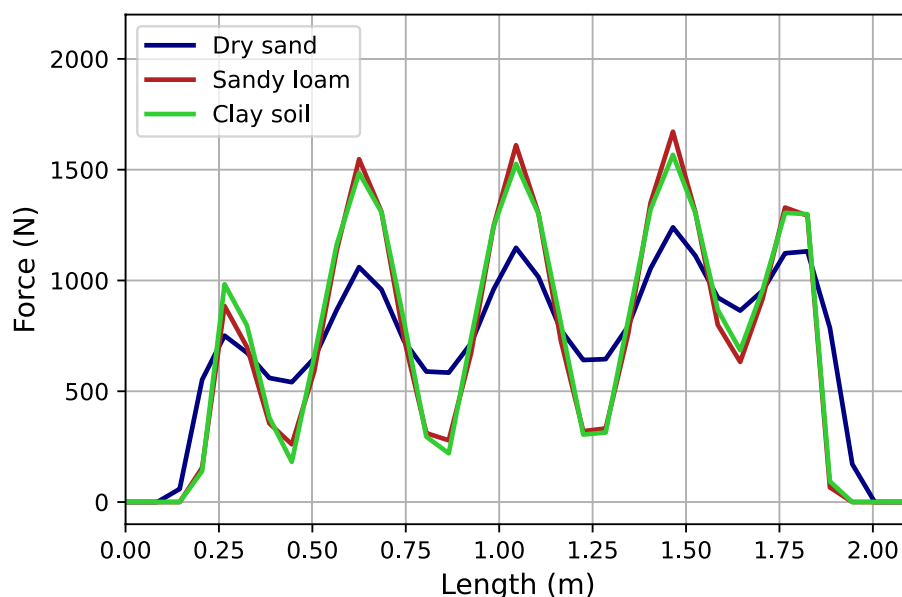


Fig. 9. Normal force distribution under the crawler.

Vehicle at rest

Initially, the crawler vehicle is simulated at rest to analyze the normal distribution along the length of the crawler. This simulation was carried out for three types of soil and vertical load track distribution on the soil, and the result is shown in Figure 9.

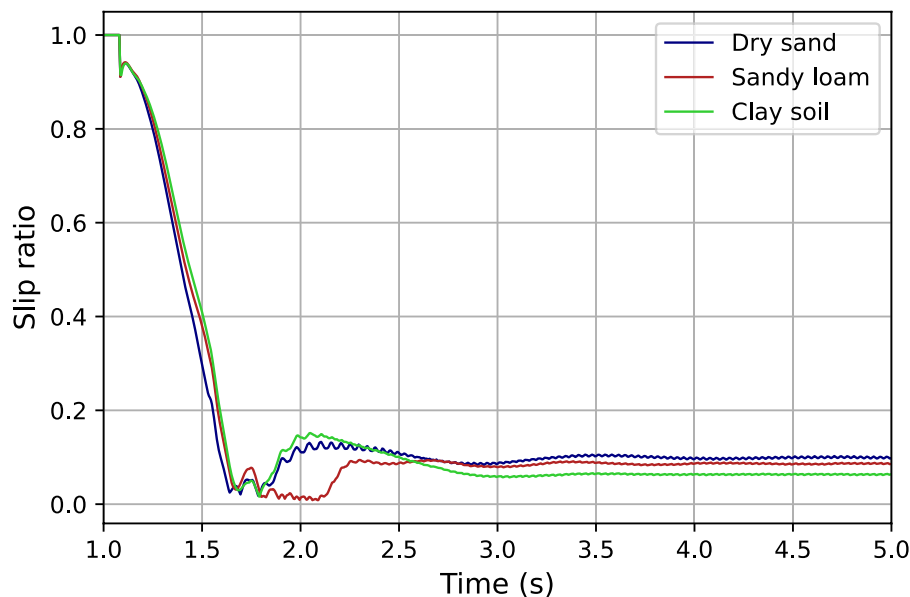
The result shown in Figure 9 represents the contact length compared to the total length of the track. The vehicle has a track that is 2.1 meters long, extending from the Sprocket wheel's edge to the edge of the idler wheel. Also in Figure 9, it can be noted that, for dry sand soil, there is an extra number of track segments in contact with the soil, which means that the vehicle sinkage is high. The center of gravity is slightly dislocated from the center, which make the vertical forces higher in the front. This result also shows three high points representing the three bottom rollers. the tracks near the idle wheel are in contact due to the sinkage in that part, pitching the vehicle in the front. Due to the geometry, the sprocket and idler wheels have vertical load components, especially for the sandy loam and the clay soil, but less than the bottom roller components.

With this simulation, it is also possible to determine the motion resistance generated by the soil sinkage. This value can be determined using Equation (16). The Table 5 shows the calculated values for the motion resistance for every soil under study. Table 5 shows that some soil leads to higher resistance, but these account for at most 2% of the vehicle's weight.

Vehicle in motion without draft load

The simulation is performed with controlled angular velocity of the sprockets, as described earlier, the sprocket wheels are activated, and a stable speed is reached. The vehicle sprockets start to move at 1 s, the velocity

Soil parameter [S.I]	Dry sand	Sandy loam	Clay soil
Motion resistance for track [N]	491.67	194.05	461.45

Table 5. Motion resistance calculated for three soils.**Fig. 10.** Slip ration of crawler vehicle with no draft load.

transition lasts 1 s until the vehicle reaches 7 km/h. To prevent instabilities in the simulation, the slip ratio is initialized at 1. Therefore, in the initial moments, the track generates a greater shear force to propel itself. The MBS model without draft load reaches a slip ratio near 9.8% dry sand, 8.7% sandy loam, and 6.2% clay soil, respectively. Figure 10 presents the slip ratio for every soil when the vehicle reaches 7 km/h.

Another aspect that is simulated and could be examined is the shear force for standard track link variation through time, for the three types of soil, which is shown in Figure 11. For the figure, segment 1 was analyzed, starting its position at the lower part of the sprocket wheel. When the segment makes contact with the soil at 3.3 seconds, the vehicle has already reached a steady speed close to 7 km/h. For the same segment, with the vehicle at a speed of 7 km/h, the different soil types affect the shear force in distinct ways. The highest magnitudes were reached with sandy loam soil, followed by dry sand, while clay soil exhibited the lowest magnitude.

The result shows an initial force during the transition from the beginning of the movement, and then when the track completes its first cycle, it shows five high shear force points that can be interpreted as the bottom roller contact pass, the sprocket, and idle wheel contact pass.

Vehicle in motion with the draft load

First, we consider the three soil types with a horizontal load due to a planter draft. The MBS model and the coefficient of slip are recorded for each soil, with a draft load close to 13 kN. Figure 12 represents the variation of the slip coefficients for each soil condition.

Considering what is shown in Figure 12, we can infer that the difference between the soils, given the speed difference between the vehicle and the driving wheel, will result in a significant variation in both vertical and horizontal loads on the vehicle. The slip coefficient for clay soil has a higher difference than the other soil types, nearly 15%. This suggests greater difficulty in achieving speed, considering the draft load of the planter. On the other hand, the sandy loam soil presents a lower slip ratio, close to 11%, which provides an advantage for the traction vehicle and enhances efficiency in the drive wheels. Finally, the slip ratio is approximately 12.45% when the vehicle travels over dry sand. This slip ratio is connected to the shear force, limiting the vehicle's drawbar pull force. Consequently, the shear force diminishes when the vehicle moves over clay soil due to its dependency on the contact area more than vehicle weight. These findings align with existing literature, which reports similar values^{11–13}. Figure 13 resumes the slip ratio with and without the draft force.

Vehicle moving in ramps

As part of the second scenario, the vehicle is simulated, climbing an inclined terrain. In this simulation, first, the vehicle is expected to reach steady-state velocity, and then it begins to contact the inclined terrain. Figure 14 represents the slip ratio during the movement. The slip ratio variation indicates a change in the longitudinal force due to the weight component during the climb of the slope.

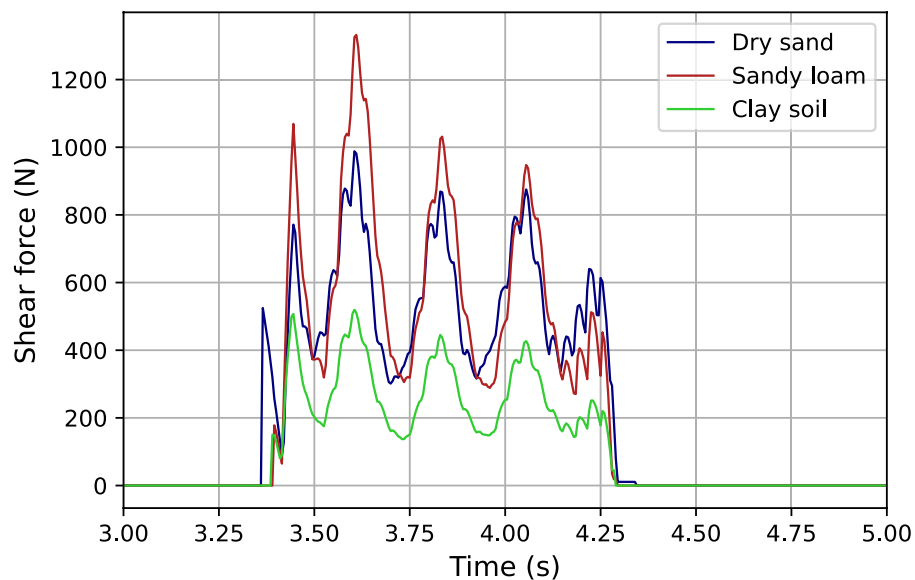


Fig. 11. Shear force during the simulation with dry sand on track link 1.

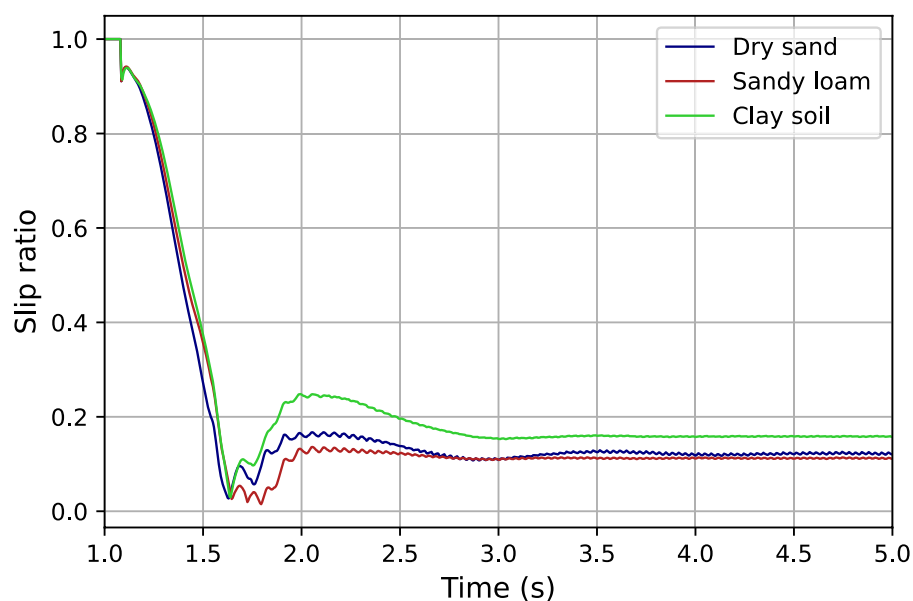


Fig. 12. Slip ratio for different soil conditions during the first scenario.

These changes require the vehicle to have a greater drawbar pull in order to reach the required speed on flat terrain. This is also reflected in the slippage ratio, which increases, as observed in Figure 14b. During the transition, an increase is also observed, even though some of the vehicle's links were not in contact with any part of the terrain. After this, the slip ratio increases from 10% to 20%.

Rubber belt tension results

In Figure 15, we observe the variation in longitudinal tension over time for vehicle simulations with and without the draft load, for each type of soil at a speed of 7 km/h. The longitudinal tension was measured by calculating the force between the rubber belt segments. This allows us to obtain the force value throughout the entire belt path.

The crawler belt tension varies depending on where the belt segment is located. Thus, the belt tension is relatively low on the upper part of the track and increases when the track element contacts the bottom rollers, idler, and sprocket wheel. In Figure 15, the graph displays seven key points of change in the longitudinal tension of the track. These points correspond to interactions with the internal components of the track. At point 1, the first change is calculated, occurring precisely when the tension between the segment joints reaches the idler wheel. At point 2, the joint reaches the lower part of the idler wheel. Points 3, 4, and 5 show an increase due

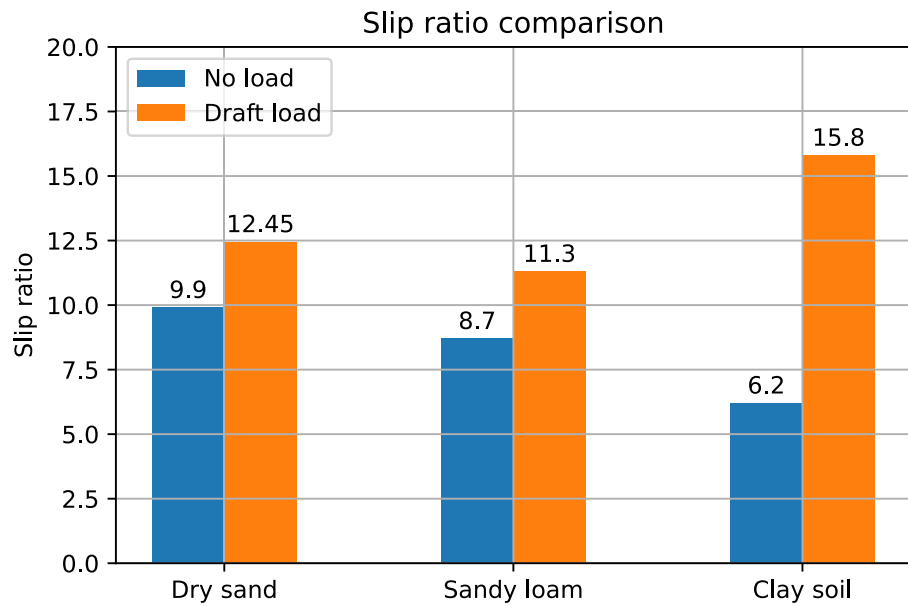


Fig. 13. Slip ratio for different soil conditions during the first scenario.

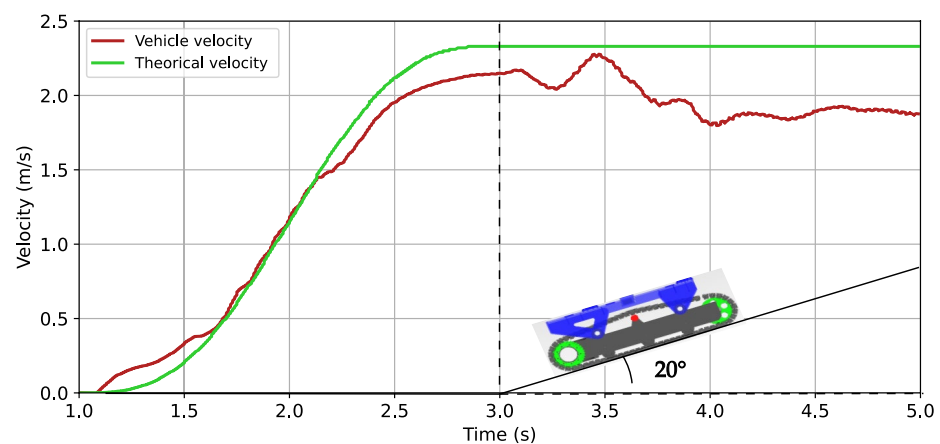
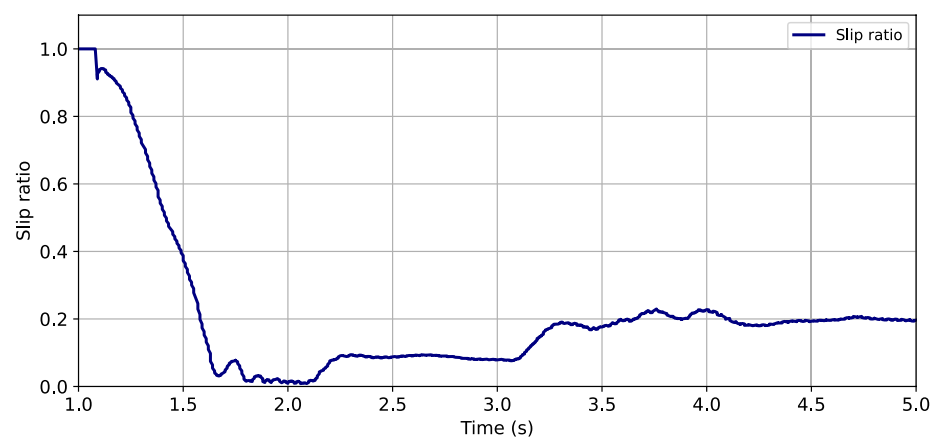


Fig. 14. (a) vehicle slip ratio in the second scenario, (b) vehicle velocity comparison.

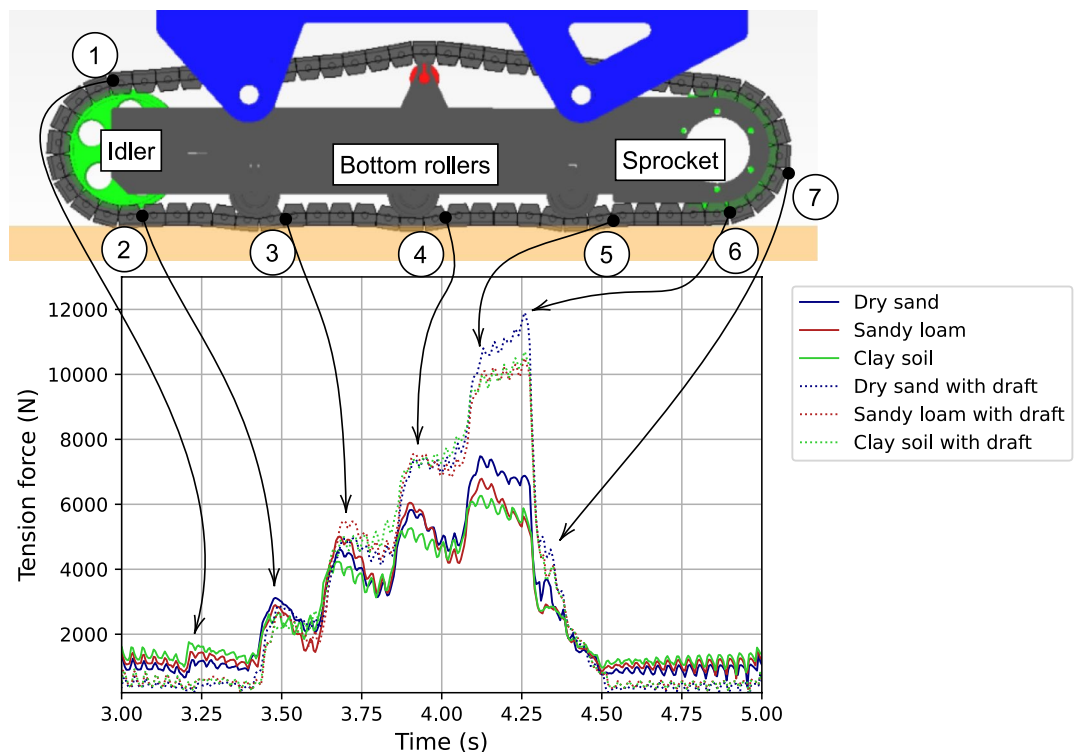


Fig. 15. Tension between segments cycle at 7 km/h for different types of soil.

to interaction after passing through the bottom roller bearings. At point 6, the highest tension is recorded, coinciding with the lower part of the sprocket wheel. Finally, at point 7, the tension between the joints gradually decreases as it passes through the sprocket wheels, returning to the initial tension observed before point 1. When the soil effect is incorporated, the belt experiences greater tension in the lower bearings, especially in dry sand, as it undergoes greater sinkage and generates more significant deflection in the belt. The tension reached its maximum at the sprocket, which is around 7.5 kN without the draft load and close to 12 kN with it. On the other hand, clay soils have less sinking, which results in lower tension values, which at its maximum is close to 6.2 kN without draft and 10.5 kN with it. When the draft load is added, the tension values increase in almost the same locations. Again, the difference between soils is significant. The tension increases to 60% for dry sand, 57% for loam sand, and 77% for clay soil of the peak without draft load.

Discussions about the results

The premise of this study is to develop a multibody model of a tracked tractor vehicle for planting and analyze its performance on different terrains. For this, it was necessary to simulate both the track's internal and external contact conditions. For internal contact, PCM was used, which allows for analysis using the geometry of realistic components. Friction is another aspect considered within this contact, with a Coulomb model used for the bearing and idler contacts with the track, using a coefficient of 0.3 for both. Additionally, the stiffness of the idler wheel spring was considered, and the model includes a preload of 4 kN in each line, ensuring belt tension. For the track, a bushing model was used, considering stiffness and damping for longitudinal and angular directions, restricting other movements. This allows the track's behavior to resemble that of a rubber track presented in other studies². External forces were based on Bekker's equations for vertical forces and the Janosi-Hanamoto equation for shear forces for three different soil types. The shear force depends on the slip ratio, which is iterated along with the entire model. The results show a significant difference between the three soil types, both with and without drag load, especially in the slip ratio. According to the results, clay soil exhibits higher slip (15%) in operation with drag load. This is because this soil type depends much more on the contact area than on the vertical components of force in the soil. Such soil usually requires the consideration of studs to increase the contact area and improve vehicle traction, as reported in studies of tracked submarine vehicles²⁷. On the other hand, dry or wet sandy soil can generate a lower slip ratio in vehicle operation (between 11% and 12%). Another factor to be considered in these two soil types is motion resistance, which depends on the vehicle pressure distribution on the soil. This aspect can be significant depending on soil conditions, being also calculated and incorporated into the proposed model. These results are relevant because they can help define the vehicle's operational stability. Another tested scenario was the vehicle climbing a 20° incline. This adds part of the weight as a force opposing the movement. The simulation was carried out on sandy loam and showed a slip ratio of 20% after the vehicle stabilized on the inclined plane. The model's response presents a transition period where there is little contact increase before stabilizing at the mentioned slip value. This result agrees with those from similar MBS models, validating the model's response with similar configurations¹³. Another important aspect is that

the model provides significant information about track tension. This tension is also altered depending on soil type and whether the vehicle has a drag load. The results show the peak tension that occurs between the lower bearings. The highest peak is between the drive wheel and the adjacent bearing in dry sand. Clay and sandy loam soils showed similar peak tensions, which could result from the low motion resistance in sandy loam soil and the low load generated by clay soil. Studies like this help complement the effects of drag loads on vehicles, which can be crucial for design, optimization, and stability, among others, and which were not discussed by other authors with similar MBS models^{2,8,9,11,15}. Besides, there is a trend in both the slip ratio and belt tensions, as the most significant variation in tension occurs in clay soil, which has the highest increase in the slip ratio. On the other hand, dry sand and sandy loam soil have similar increases, with dry sand showing higher values in both tension and slip ratio. This highlights the importance and correlation of these two parameters and can offer an insight about the tendency for the tension behavior under specific slip conditions and soil types. The proposed model was configured to evaluate the longitudinal dynamics of a rubber-tracked vehicle. However, the model is not limited to this type of dynamic; it can also be used to analyze the dynamics of lateral forces, which will indeed be part of subsequent studies. Finally, the uniqueness of our problem and approach make it difficult to find literature results to be compared with ours. In Sandu, Dhir and Sankar^{10,28}, a crawler vehicle is analyzed using numeric model. The authors show a distribution of tension that is very similar to the one we presented in Figure 15. The higher magnitude of those results can be explained by the stiffness and design she used. In Grazioso et al.¹¹, the authors show the distribution of the normal forces, which is quite like the distribution that we showed in Figure 9. The differences are mostly in the magnitude of the forces and due to design since their crawler has different sizes and setup. Other researchers analyzed different aspects and their results are not adequate to be compare with ours^{15,18,29}.

Conclusions

In this paper, an MBS model of a rubber crawler vehicle for agriculture is proposed. The MBS was configured under three types of soil conditions and slope climb conditions, and variables like the vertical load, the tension of the rubber belt, and the longitudinal load were examined. The MBS model was configured for three scenarios: the plane terrain of 3 types of soil, a slope climbing for dry sand soil type, and plane terrain with three types of soil and draft load representing a planter and seeding operation, with a standardized draft load. Initially, the MBS results presented the vertical load distribution due to the Center of Mass location showing a higher force in the front part, especially on the bottom rollers. This is also related to the belt track tension that could alter the distribution in the tracks. The initial results also showed the slip ratio of the vehicle with no draft load in a horizontal plane. Additionally, the motion resistance was evaluated using the vehicle's pressure distribution on the soil, with the maximum resistance observed in dry sand soil and the minimum in sandy loam. When the vehicle drives over the dry sand, the sandy loam and clay soil reach different slip ratio with maximum for the dry sand and minimum for clay soil. The MBS result was analyzed using parameters like the slip ratio, vehicle velocity during, vertical loads on the tracks, and belt track tension. Based on the simulation results, there is an increased slip ratio due to the draft load related to the plant and seeding operation, which is higher for the clay soil and lower for the sandy loam. These results are supported by the sinkage of the vehicle in each type of soil and the shear force generated during the drive. Additionally, the values found are confirmed by literature^{10–13,28}. Conversely, when the vehicle climbs a 20° incline plane with respect to the horizontal, there is an interval of transition between the horizontal plane and the slope plane. In part of the simulation, the vehicle increases the slip ratio due to the addition of weight force components, reducing the vehicle speed at the same rotational speed in the sprocket. However, the slip ratio increases when the vehicle climbs, reaching 20 %. The model presented that the maximum tension reaches up to three times the average tension when the track element is near the sprocket. When planter reaches it maximum speed the effect of the soil is more pronounced, which is more evident when the segment of the belt contacts the bottom of the rollers. The magnitudes of the tensions become even higher under dry sand conditions. During the passage of the lower rollers, the internal force is lower in sandy loam soil and clay. The results also present a correlation between the increase in slip ratio and the tension due to the draft load. Clay soil increases the most in both the slip ratio and belt tension. In the same way, the slip ratio and belt tension increase in a similar proportion for sandy loam and dry soil. The results of this study used the effect of soil terramechanics on the rigid MBS of crawler vehicles. The findings were consistent with data reported by previous researchers^{12,30}. Finally, the analysis of the slippage and the load during the different scenarios presented important insights into the crawler's internal rolling dynamics, allowing future development of even more complex's models for agricultural vehicles.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to being part of a larger study to develop a new agricultural vehicle, which is not completed by now, but are available from the corresponding author on reasonable request.

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Author contributions

R.J.C.C.: Writing – original draft, Software, Resources, Methodology, Investigation, Methodology, Conceptualization. W.L.A.: Writing – review & editing, Validation, Software, Methodology, Conceptualization. A.P.G.: Writing – review & editing, Validation, Software, Data curation, Investigation, Project administration. M.V.P.: Writing – review & editing, Validation, Resources, Project administration. A.A.S.: Writing – review & editing, Validation, Resources, Project administration.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to R.J.C.C.

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